

tac_plus Pre-Authentication Format-String Vulnerability (Advisory)

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Content

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elttam-tac_plus_rce_fmtstring_advisory.txt

elttam/publications at master, path writeups/elttam-tac_plus_rce_fmtstring_advisory.txt.

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Security Advisory

elttam

Topic: Remote pre-authentication format **string** vulnerability in
tac_plus send_authen_error() (CWE-134)

Module: packet.c - send_authen_error() / get_authen_continue()

CVE: <CVE-PENDING>

Severity: Critical - CVSS v3.1 9.8
(AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H, elttam assessment)
9.8 applies where no shared secret is in effect, or where
the configured secret is weak. Where the secret is long,
random and unique, AC:H is the better fit, giving 8.1 High.
See section III.

Disclosed: 23-Sep-2026

Credits: Matt Jones, elttam

Affects: Shrubbery Networks tac_plus, up to and including F4.0.4.31
(Feb-2026). Both sinks described below are present in that
release. Fixed in F4.0.4.32, published 21-Sep-2026.

Note that F4.0.4.31 was the first official release since
F4.0.4.28 (06-Jan-2015). Through that eleven-year interval
distributions and downstream maintainers carried their own
patches against F4.0.4.28, so a given installation may not
correspond to any upstream release. Version strings alone
are not a reliable test; check **for** the sinks in the source
actually built.

Facebook tac_plus, F4.0.4.28-7fb. The send_authen_error()
sink is present; get_authen_continue() in **this** fork already
passes a constant format **string** and is not affected. The
repository was archived on 22-Aug-2025 and is read-only.

Other Cisco-derived forks carrying the same report() and
send_authen_error() code should be assumed affected. The
only actively packaged copies found are the FreeBSD port
net/tac_plus4 and the pkgsrc entries net/tacacs-shrubbery
and net/tacacs. Debian removed its tacacs+ **package** in
September 2019 at 4.0.4.27a-3, Ubuntu carries it in
universe on older LTS suites only, and Arch ships it only

in the AUR.

"Cisco-derived" refers to the historical open-source TACACS+ reference server published by Cisco, **from** which these third-party forks descend.

Cisco PSIRT has confirmed that no Cisco product uses the Shrubbery or Facebook forks, or Cisco's own original version of that code, and that no Cisco product is affected.

Note: Line numbers in **this** advisory refer to the Shrubbery F4.0.4.31 source tree. Offsets differ in other forks; in F4.0.4.28-7fb the `send_authen_error()` sink is at approximately `packet.c:223`.

I. Background

`tac_plus` implements the TACACS+ AAA protocol used by network devices **for** administrative login and command authorization. It listens on TCP port **49** and typically runs as root.

Errors are logged through a printf-style helper, `report(int priority, char *fmt, ...)` (`report.c:51`), which passes `fmt` directly to `vsnprintf()` (`report.c:70`). Any conversion specifiers in `fmt` are therefore expanded against the variadic arguments.

II. Problem Description

`send_authen_error()` builds a message **from** caller and session data, then passes it to `report()` as the format **string** rather than as a data argument:

```
packet.c:235  snprintf(buf, sizeof(buf), "%s %s: %s", session.peer,
               session.port, msg);
packet.c:238  report(LOG_ERR, buf);           /* buf is the format string */
```

`buf` embeds `session.port`, which is attacker-controlled: it is copied verbatim **from** the **"port"** field of the client's AUTHEN/START request with no validation or character restriction (`authen.c:108-114`; the same applies to AUTHOR and ACCT requests). A client that places conversion specifiers such as `%x` or `%n` in that field has them interpreted by `report()`'s `vsnprintf()` against arguments that were never supplied. When peer-name lookups are enabled (`-L`), the reverse-DNS name in `session.peer` is a second tainted source in the same buffer.

The same mistake appears in `get_authen_continue()` in the Shrubbery tree, which passes a message containing `session.peer` straight to `report()` as the format

string:

```
packet.c:61    report(LOG_ERR, msg);           /* msg is the format string */
```

III. Impact

This is a remote, pre-authentication format **string** vulnerability in a service that runs as root by **default**; privileges are dropped only when the daemon is invoked with -U or -Q. The %x and %s specifiers disclose process memory into the daemon's log, and %n writes to memory, enabling memory corruption and remote code execution as the daemon user.

report() expands the format **string** into a buffer of its own and does not modify buf, so the AUTHEN reply sent by send_authen_reply() carries the specifiers verbatim rather than expanded.

There is no read-back channel to the client, and the write primitive is therefore blind. **This** has been confirmed against a running daemon.

elttam has confirmed remote code execution against a build with ASLR and NX disabled. Exploitation of a hardened build carrying the full set of mitigations is believed to be achievable, since the daemon forks a child per connection without exec(), so address layout is stable **for** the life of the process and a failed attempt costs only that child. Compromise of the daemon can cascade into control of the network devices that rely on it **for** administrative access.

No user credentials are required: the vulnerable path is reached before authentication. Exploitation does, however, depend on the TACACS+ obfuscation layer:

- * Where no shared secret is configured, or the client sets the unencrypted flag and the daemon accepts it, an attacker needs only network reach to TCP/49 to place a chosen payload in the port field.
- * Where a shared secret is configured, the request body is obfuscated with a key the daemon must be able to reproduce, so delivering a chosen payload requires knowledge of that key.

How much protection that key provides depends on its quality rather than its presence. send_authen_error() returns a reply before the peer has proven any knowledge of the key, and the plaintext of that reply is predictable to the client that requested it, which permits offline verification of candidate keys **from** a single connection with no captured traffic and no on-path position. elttam has confirmed **this** in practice against a weak key. **This**

behaviour is a property of the TACACS+ obfuscation scheme, described by Solar Designer in 2000 and superseded by RFC 9887, rather than a separate defect in tac_plus; it is not the subject of [this](#) advisory, has no CVE assigned, and was not separately reported.

The two ratings above follow [from](#) that. Where no secret is in effect, or the secret is weak, guessable or shared with other systems, the attacker needs only network reach to TCP/49, giving 9.8. Where the secret is long, random and unique, AC:H is the better fit, giving 8.1.

IV. Solution

Shrubbery Networks responded on 01-Sep-2026, after the report was recovered [from](#) a mailing list spam queue, and published F4.0.4.32 on 21-Sep-2026 with a fix [for](#) both sinks. elttam has verified that release: both calls now pass a constant format [string](#), at packet.c:61 and packet.c:238, the same line numbers as in F4.0.4.31. Operators running the Shrubbery fork should upgrade to that release. Note that F4.0.4.32 is now the only version available in Shrubbery's download directory; the earlier tarballs have been removed.

The Facebook repository was archived on 22-Aug-2025 and is read-only, so no fix will be issued there. Operators of that fork, of other Cisco-derived forks, or of distribution packages that have not yet picked up the fixed release, should assume they must patch their own builds, or migrate to another implementation.

Pass a constant format [string](#) and supply the data as an argument at both sinks:

```
packet.c:238    report(LOG_ERR, buf);    ->    report(LOG_ERR, "%s", buf);
packet.c:61     report(LOG_ERR, msg);    ->    report(LOG_ERR, "%s", msg);
```

The strategic fix is to stop relying on the obfuscation layer at all.

RFC 9887 (December 2025, Standards Track) specifies TACACS+ over TLS 1.3 and obsoletes obfuscation, stating that [new](#) production deployments SHOULD [use](#) TLS authentication and encryption. Operators with a migration path to a TLS-capable implementation should take it.

More broadly, audit every report(), syslog() and *printf() call so that the format argument is always a [string](#) literal and never runtime data. Building with -Wformat -Wformat-security -Werror=format-security flags the remaining cases at compile time. Request-derived values - the NAS port field, and DNS-derived peer names when -L is used - should be treated as untrusted input.

V. Workarounds

Operators who cannot rebuild the daemon should consider the following. None of these fully remediate the vulnerability.

- * Restrict TCP/49 to trusted NAS management addresses at the network layer.
- * **Use** a long, random, unique shared secret. Any secret a wordlist or rule-based attack can reach is equivalent to no secret at all against the oracle in section III. **This** does not prevent exploitation by a party who already knows the key.
- * Avoid -L, which removes the reverse-DNS peer name as a tainted source. The NAS port field remains attacker-controlled regardless.

elttam does not recommend -U/-Q as a mitigation. The flags work as documented, with the daemon binding its listening socket before it calls `setgid()` and `setuid()`, but they **do** not prevent exploitation, and the shared secret and the credentials the daemon handles are readable by the daemon's own account whatever its uid.

VI. Detection

Exploitation attempts may appear in the daemon's logs as entries whose port or peer field contains expanded conversion specifiers, or as truncated or malformed `report()` output. Note that the expansion appears in the log only, not in the reply sent to the client. A successful `%n` write may occur without producing a distinctive log entry, so absence of these indicators is not evidence that a host was not targeted.

VII. Timeline

- * **15-Jan-2026**: Initial vulnerability discovery and proof-of-concept development
- * **06-Feb-2026**: Reported to the Meta bug bounty program
- * **Feb-2026**: Shrubbery published F4.0.4.31, its first official release since **06-Jan-2015**. Not noticed by elttam at the time.
- * **15-Apr-2026**: Contacted the Meta bug bounty program lead after no response
- * **15-Apr-2026**: Meta responded privately, out of band **from** its bug bounty program, directing elttam to Shrubbery Networks and to Cisco's Bugcrowd program
- * **03-Jun-2026**: Reported to Shrubbery Networks via `info@shrubbery.net`
- * **24-Jun-2026**: Follow-up to Shrubbery after no response
- * **15-Jul-2026**: Follow-up to Shrubbery after no response
- * **17-Jul-2026**: Notified the Australian Cyber Security Centre (ACSC)

- * **07-Aug-2026**: Follow-up to Shrubbery regarding the **90-day** disclosure deadline, counted **from** the **03-Jun-2026** report
- * **08-Aug-2026**: Notified Cisco PSIRT via psirt@cisco.com
- * **28-Aug-2026**: Follow-up to Shrubbery, including tac_plus@shrubbery.net
- * **31-Aug-2026**: Cisco PSIRT responded, advising that no Cisco product uses the Shrubbery or Facebook forks, or Cisco's original code
- * **31-Aug-2026**: CVE identifier requested by elttam **from** MITRE (<CVE-PENDING>)
- * **01-Sep-2026**: Shrubbery Networks responded, exactly **90** days after the **03-Jun-2026** report and on the date elttam was prepared to publish, advising that the report had been held in the mailing list's spam queue, and requesting a non-video demonstration of the issue
- * **01-Sep-2026**: elttam provided the advisory and the CVE identifier, and deferred disclosure by a week, **from** **01-Sep-2026** to **07-Sep-2026**, asking Shrubbery to confirm by **04-Sep-2026** whether a patch and release will be issued
- * **07-Sep-2026**: Follow-up to Shrubbery after no confirmation was received by the **04-Sep-2026** deadline
- * **09-Sep-2026**: Shrubbery advised that it would be addressed on **return** to the office on **14-Sep-2026**, with nobody available to publish a release before then. Disclosure deferred a second time, **from** **07-Sep-2026** to **15-Sep-2026**.
- * **13-Sep-2026**: elttam asked Shrubbery to confirm the release and version number once published
- * **16-Sep-2026**: Follow-up to Shrubbery after **14-Sep-2026** passed with no release and no further response
- * **17-Sep-2026**: Further follow-up to Shrubbery, advising that elttam is ready to publish and has held off to see whether a patched release is forthcoming
- * **21-Sep-2026**: Shrubbery published F4.0.4.32 with a fix **for** both sinks
- * **23-Sep-2026**: **Public** disclosure of advisory

The report was submitted to the Meta bug bounty program on **06-Feb-2026** and received no substantive response through that channel. On escalation to the program lead on **15-Apr-2026** the submission was located and passed to an analyst, with the qualification that scope covers active GitHub projects and **this** one was deprecated, making an end-of-life CVE the likelier outcome than a bounty. elttam was also directed to Shrubbery Networks and to Cisco's Bugcrowd program. No end-of-life CVE was issued and the analyst review produced no further response, so elttam requested the identifier **from** MITRE directly.

elttam approached Meta first because, at the time of discovery, Shrubbery's

most recent official release was F4.0.4.28 from January 2015, whereas the Facebook fork had been maintained publicly until its archival in August 2025, making Meta the party most likely to retain familiarity with the code and to be positioned to coordinate with others running it. Meta's bug bounty program treats the company's active GitHub projects as in scope, and the fork sat within that scope from late 2015 until its archival in August 2025, some months before elttam reported.

Following that referral, and before contacting Shrubbery Networks, elttam attempted to report the issue through Cisco's Bugcrowd program. The intake form accepted a basic request but not a full report, and retained no submission record or date, so the attempt is not listed in the timeline above. The issue was reported directly to psirt@cisco.com on 08-Aug-2026. Cisco PSIRT replied on 31-Aug-2026, noting that the original message had been caught as spam, and advised that no Cisco product uses the Shrubbery or Facebook forks, or Cisco's own original version of that code, and that the Developer's Kit was not intended for production use.

Shrubbery Networks did not respond to the initial report or to the four follow-ups sent between 03-Jun-2026 and 28-Aug-2026, and replied on 01-Sep-2026 advising that the correspondence had been held in a mailing list spam queue. Disclosure was deferred twice from that point, the second time to 15-Sep-2026 after Shrubbery advised that the issue would be addressed on its return to the office on 14-Sep-2026. That date passed without a release, and follow-ups on 13-Sep-2026, 16-Sep-2026 and 17-Sep-2026 received no response. F4.0.4.32 was published on 21-Sep-2026, recorded in CHANGES as a single line naming the bug class, without the CVE identifier or any credit. No fix is available for the archived Facebook fork.

VIII. References

- * CWE-134: Use of Externally-Controlled Format String
<https://cwe.mitre.org/data/definitions/134.html>
- * RFC 9887: TACACS+ over TLS 1.3 (Standards Track, December 2025)
<https://www.rfc-editor.org/rfc/rfc9887.html>
- * Shrubbery Networks tac_plus: https://www.shrubbery.net/tac_plus/
- * Facebook tac_plus (archived): https://github.com/facebook/tac_plus
- * An Analysis of TACACS+ Protocol Security, Solar Designer, Openwall, 2000
<https://www.openwall.com/articles/TACACS+-Protocol-Security>
(documents offline recovery of the key from captured traffic; the oracle in section III removes the need to capture any)
- * Affected source: packet.c (send_authen_error at packet.c:238, get_authen_continue at packet.c:61, md5_xor, read_packet, write_packet), report.c (report), authen.c:108-114 (session.port from the NAS port

```
field), authen.c:authen (reaches send_authen_error pre-authentication)
* Affected version string: version.h.in -> version[] = "F4.0.4.31"
* Fixed release: tacacs-F4.0.4.32.tar.gz, published 21-Sep-2026,
MD5 d380e9005363f4e9f25bf1b40cf4cab7,
SHA256 ee3e403a079c75cad13ce9e451c1af0115b0647142765bbd85e60d7a9f3f3fd2
```

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